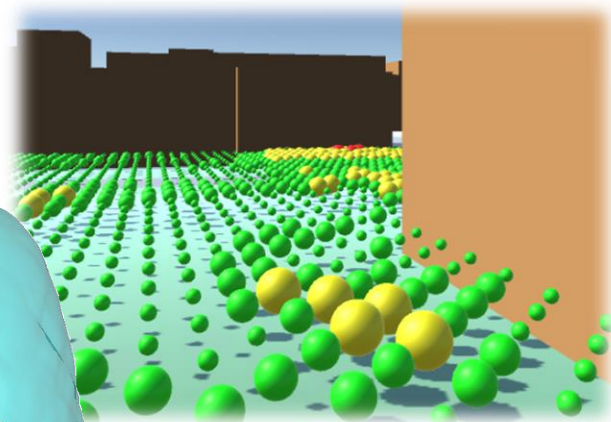
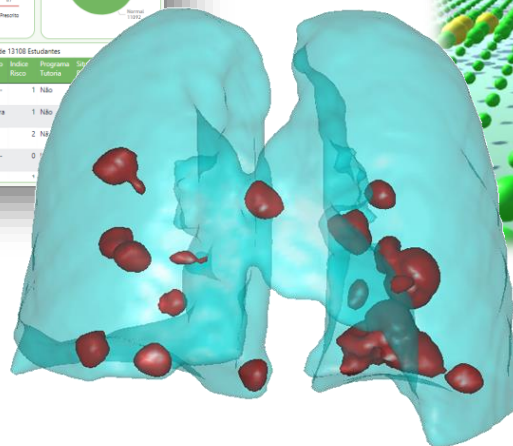




# Data Visualization module 2025

## Introduction



# What is Data Visualization?

- Visualization is focused on how to **visually represent and explore large amounts of data**
- Taking advantage of the **human visual system capacities**
- Providing “**insights**” concerning the phenomenon behind the data

## What it **is not**:

- just “pretty pictures”!

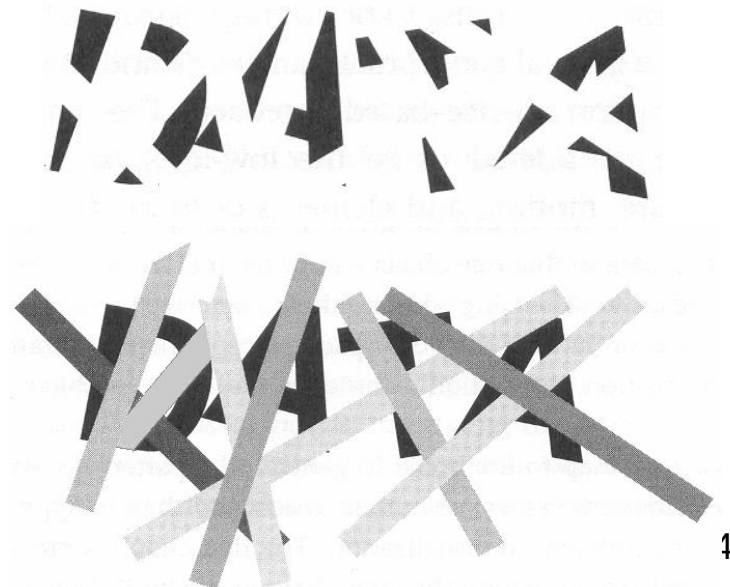
# Why and how to represent data visually?

- The human visual system is a most powerful pattern seeker

“seeing is understanding...”

- We easily see patterns displayed in certain ways

but not in others ...



An exercise in preattentive processing: how many “3”?

69704259347493  
58728294954642  
44396854634235  
6658789376

(Nussbaumer Knaflitz, 2015)

69704259347493  
58728294954642  
44396854634235  
6658789376

It is much easier to perform this task!

What if: How many “7”?

C. Nussbaumer Knaflic, Storytelling with Data ,Talks at Google, 2015  
[Storytelling with Data | Cole Nussbaumer Knaflic | Talks at Google](#)

# Why represent data visually?

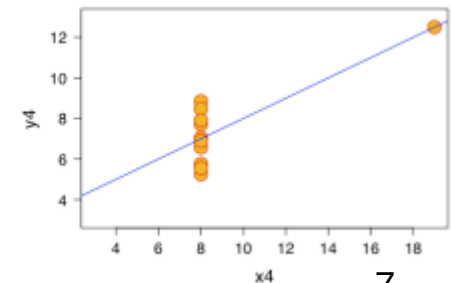
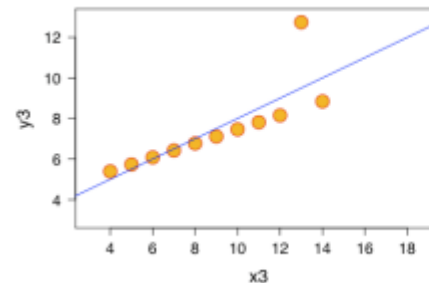
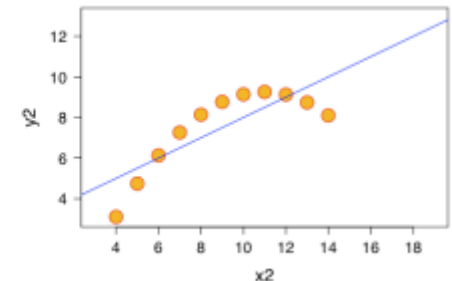
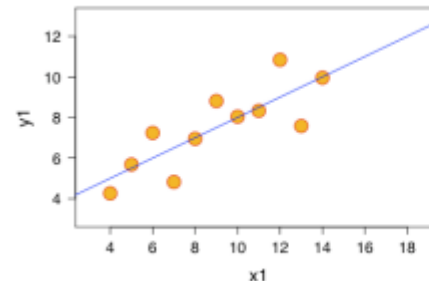
Vis helps in situations where seeing the dataset structure in detail is better than seeing only a brief summary of it.

(Munzner, 2014)

| Anscombe's Quartet: Raw Data |       |      |       |      |       |      |       |      |
|------------------------------|-------|------|-------|------|-------|------|-------|------|
| I                            |       | II   |       | III  |       | IV   |       |      |
| x                            | y     | x    | y     | x    | y     | x    | y     |      |
| 10.0                         | 8.04  | 10.0 | 9.14  | 10.0 | 7.46  | 8.0  | 6.58  |      |
| 8.0                          | 6.95  | 8.0  | 8.14  | 8.0  | 6.77  | 8.0  | 5.76  |      |
| 13.0                         | 7.58  | 13.0 | 8.74  | 13.0 | 12.74 | 8.0  | 7.71  |      |
| 9.0                          | 8.81  | 9.0  | 8.77  | 9.0  | 7.11  | 8.0  | 8.84  |      |
| 11.0                         | 8.33  | 11.0 | 9.26  | 11.0 | 7.81  | 8.0  | 8.47  |      |
| 14.0                         | 9.96  | 14.0 | 8.10  | 14.0 | 8.84  | 8.0  | 7.04  |      |
| 6.0                          | 7.24  | 6.0  | 6.13  | 6.0  | 6.08  | 8.0  | 5.25  |      |
| 4.0                          | 4.26  | 4.0  | 3.10  | 4.0  | 5.39  | 19.0 | 12.50 |      |
| 12.0                         | 10.84 | 12.0 | 9.13  | 12.0 | 8.15  | 8.0  | 5.56  |      |
| 7.0                          | 4.82  | 7.0  | 7.26  | 7.0  | 6.42  | 8.0  | 7.91  |      |
| 5.0                          | 5.68  | 5.0  | 4.74  | 5.0  | 5.73  | 8.0  | 6.89  |      |
| mean                         | 9.0   | 7.5  | 9.0   | 7.5  | 9.0   | 7.5  | 9.0   | 7.5  |
| var.                         | 10.0  | 3.75 | 10.0  | 3.75 | 10.0  | 3.75 | 10.0  | 3.75 |
| corr.                        | 0.816 |      | 0.816 |      | 0.816 |      | 0.816 |      |

[Anscombe's quartet - Wikipedia](#)

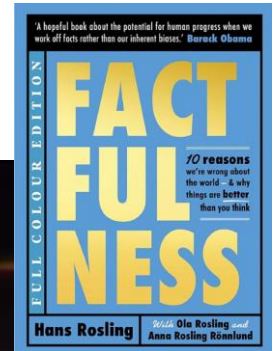
**Anscombe quartet:** data sets with the same simple statistical properties  
(Tuft, 1983)



# Example of Presentation based on a simple Visualization:

## World health by Hans Rosling

200 years of health/income – 120 000 values in 4 min



[Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats – BBC](#)  
[Factfulness Illustrated: Ten Reasons We're Wrong About the World - Why ... -](#)  
[Hans Rosling, Ola Rosling, Anna Rosling Rönnlund - Google Books](#)

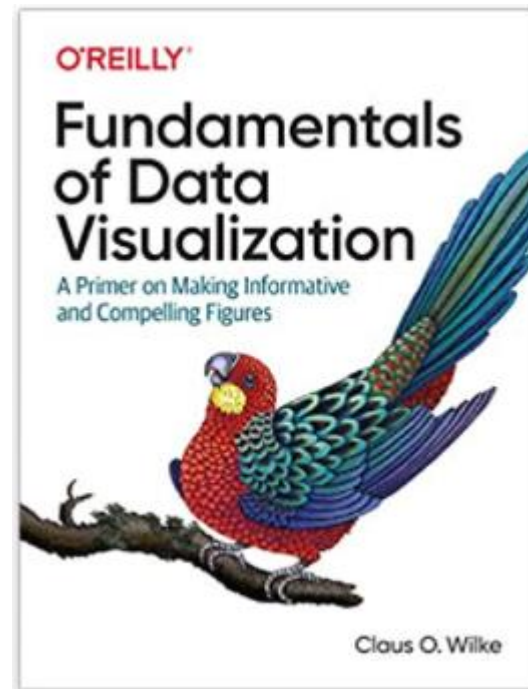
# This module:

- Objective, bibliography and tools
- Introduction to Data Visualization and its application
- Data and phenomena: types and pre-processing
- Human-in-the-loop process: perceptual and cognitive aspects
- Visualization of tabular and field data:
  - 1D and 2D Data – main methods; exercises
  - 3D and nD data - main methods; exercises
- Visualization of other types of data
  - georeferenced, networks, hierarchical data, etc.
- Effective Visualization



# Bibliography

- [Camões, J., \*Data at Work : Best practices for creating effective charts and information graphics in Microsoft Excel\*, Pearson Education, 2016](#)
- [Wilke, C., \*Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures\*, O'Reilly, 2019](#)  
[Fundamentals of Data Visualization](#)



# More advanced bibliography

- Kirk, A., Data Visualisation A Handbook for Data Driven Design, 2nd. Ed., Sage, 2019
- [Kirk, A., \*Data Visualization: A successful design process\*, Pack Publishing, 2012](#)
- [Munzner, T., \*Visualization Analysis and Design\*, CRC Press, 2014](#)
- [Telea, A., \*Data Visualization: Principles and Practice\*, CRC Press, 2015](#)
- [Ware, C., \*Information Visualization, Perception for Design\*, 3rd ed., Morgan Kaufman, 2012](#)

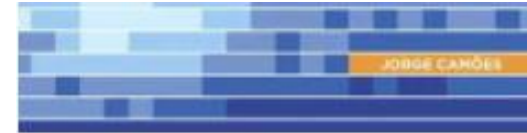
Explore yet other books available at the playlist:

<https://learning.oreilly.com/playlists/74bfec5e-4346-48ff-82b4-657fda6922b6>



# Interesting links

- [Everything by Jorge Camoes](#)



- [Eagereyes.org](#)

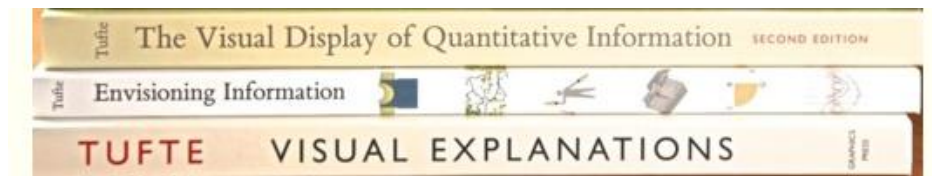


- [Perceptual Edge](#)



**Visual Business Intelligence**  
for enlightening analysis and communication

- [The Work of Edward Tufte & Graphics Press](#)



# Interesting links



- [Multiple Views: Visualization Research Explained](#)

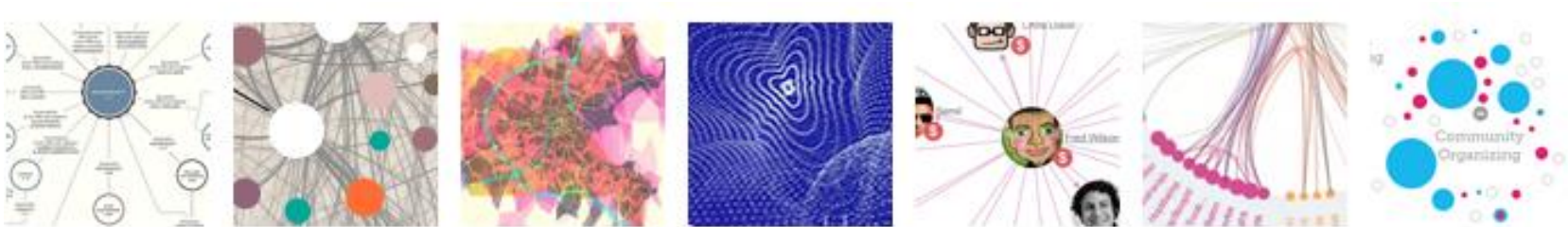
- [Home - Seeing Data](#)



- [FlowingData](#)



- [VisualComplexity.com. 10 Years / 1000 Projects | by Manuel Lima](#)



# Moodle



## Slides



Introduction to the module



Introduction to Visualization



Data



Creating a Visualization



Effective Visualization



## Bibliography



Data - Basic concepts



Creating a Visualization



Camões, J., Data at Work : Best practices for creating effective charts and information graphics in Microsoft Excel, Pearson Education, 2016



Wilke, C., Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures, O'Reilly, 2019



Healy, K., Data Visualization a Practical Introduction, Princeton, 2019



Kirk, A., Data Visualization: A successful design process, Pack Publishing, 2012



Spence, R., Design for Interaction (2nd Edition), Pearson, 2007



More Advanced Bibliography- playlist

# Sessions

(subject to minor adjustments)

- 1** - Introduction to the module. Data Visualization – overview and applications
- 2** - Visualization model. Data - different types, phenomena they represent and pre-processing. Visualization Literacy Quiz
- 3** - Creating a Visualization – questions guiding the design; visual encoding  
Visualization of 1D and 2D, 3D tabular data – main techniques; simple exercises
- 4** – Effective Visualization; application examples and critical analysis of visualizations
- 5** – Application exercises, quizzes and critical analysis of visualizations
- 6** - Visualization of multivariate tabular data and fields – main techniques; examples
- 7** – Application exercises, critical analysis of visualizations and quiz
- 8** - Visualization of relation and hierarchical data – main techniques; quiz
- 9** - Test

# **Assessment**

## **During the semester:**

Test – 100%

## **Alternative at the end of semester:**

Exam – 100%    June, 20<sup>th</sup> / July, 8<sup>th</sup>

Test – last session of the module – June, 5<sup>th</sup>

# Practical activities

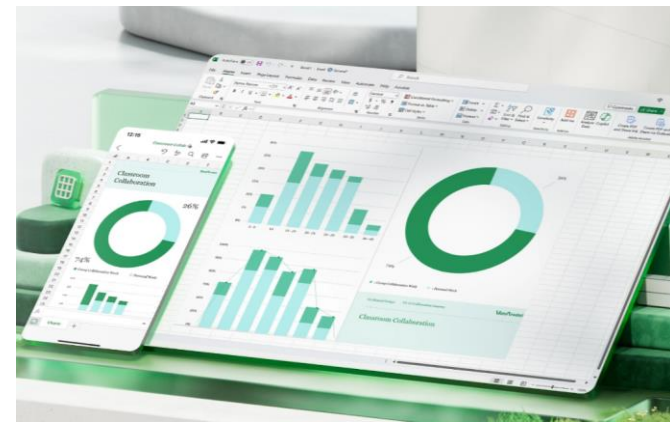
Quizzes, exercises and critical analysis

[Microsoft-365/excel](#)

[Matplotlib](#)

Next lecture bring your own laptop!

Questions regarding the module?



matplotlib

